WEEK 8

Multi-Qubit Systems, CNOT, and Qiskit

6, 8 October · Objectives 3, 4

One qubit has a sphere. Two qubits have a 6-parameter space with no three-dimensional picture, and the extra structure is entanglement. This week adds the two-qubit gate that creates it, and puts the toolchain in your hands.

Composite state spaces

Two qubits span a four-dimensional space:

```
|ψ⟩ = a|00⟩ + b|01⟩ + c|10⟩ + d|11⟩      |a|²+|b|²+|c|²+|d|² = 1
```

Convention: $|q_1 q_0\rangle$ with **qubit 0 rightmost** — Qiskit's little-endian ordering, and a reliable source of off-by-one confusion when reading bitstrings off hardware.

The separability test from Week 4, made computational: the state factors if and only if **ad − bc = 0**. That determinant is the two-qubit entanglement witness, and it is worth committing to memory:

```
|Φ⁺⟩ = (|00⟩ + |11⟩)/√2  →  a=d=1/√2, b=c=0  →  ad-bc = 1/2 ≠ 0   entangled
|+⟩|0⟩ = (|00⟩ + |10⟩)/√2 →  a=c=1/√2, b=d=0  →  ad-bc = 0        separable
```

CNOT and the entangling gates

CNOT flips the target if the control is $|1\rangle$:

```
         [1 0 0 0]
CNOT =   [0 1 0 0]       |00⟩→|00⟩    |10⟩→|11⟩
         [0 0 0 1]       |01⟩→|01⟩    |11⟩→|10⟩
         [0 0 1 0]
```

On basis states it is classical XOR. On superpositions it is the entangling engine:

```
|+⟩|0⟩ = (|00⟩+|10⟩)/√2  --CNOT-->  (|00⟩+|11⟩)/√2 = |Φ⁺⟩
```

One Hadamard and one CNOT produce a Bell state. That two-gate circuit is the "hello world" of quantum computing and the basis of teleportation, superdense coding, and every entanglement benchmark you will run on hardware.

Phase kickback, which will matter enormously in Week 11: apply CNOT with the *target* in |−⟩:

```
    |ψ⟩|-⟩  --CNOT-->  (-1)^{ψ} |ψ⟩|-⟩
```

The target is unchanged. The **control** picks up a phase. Information flows backwards relative to the gate's apparent direction — the control/target labels describe the computational basis, not a causal channel.

> **Analogy boundary — CNOT as a conditional.** It reads like `if control: flip(target)`, and on basis states that is exactly right. Two failures. First, on superpositions there is no branch taken — both branches evolve, and their relative phase persists and can later interfere. Second, phase kickback means your `if` modifies its own condition variable, which no conditional in any language does. Treat CNOT as a *linear operator on a 4-D space*, and it behaves; treat it as control flow, and phase kickback will look like magic.

Universality. CNOT plus arbitrary single-qubit rotations is universal — any n-qubit unitary decomposes into them. This is the quantum counterpart of NAND completeness, with the caveat that decompositions can be exponentially long. Efficiency is the whole game.

Qiskit

Qiskit builds circuits as objects, transpiles them for a target backend, and runs them through **primitives** — `Sampler` for measurement distributions, `Estimator` for expectation values.

```
from qiskit import QuantumCircuit

# Two qubits, two classical bits for the measurement record.
qc = QuantumCircuit(2, 2)

qc.h(0)          # qubit 0 -> |+⟩ : superposition, the entangling precondition
qc.cx(0, 1)      # control 0, target 1 -> (|00⟩ + |11⟩)/√2, a Bell state
qc.measure([0, 1], [0, 1])

print(qc.draw())
```

```
     ┌───┐     ┌─┐
q_0: ┤ H ├──■──┤M├───
     └───┘┌─┴─┐└╥┘┌─┐
q_1: ─────┤ X ├─╫─┤M├
          └───┘ ║ └╥┘
c: 2/═══════════╩══╩═
                0  1
```

Run it on the local simulator:

```
from qiskit_aer import AerSimulator
from qiskit import transpile

sim = AerSimulator()
result = sim.run(transpile(qc, sim), shots=4096).result()
print(result.get_counts())
# {'00': 2043, '11': 2053}   — never '01' or '10'
```

The absence of `01` and `11` ... rather, of `01` and `10`, is the entanglement. The two qubits always agree, and neither individually is predictable.

Inspect the state before measurement — the diagnostic you will use constantly:

```
from qiskit.quantum_info import Statevector, partial_trace, entropy

psi = Statevector(QuantumCircuit(2).compose(qc.remove_final_measurements(inplace=False)))
# Equivalent and cleaner:
bell = QuantumCircuit(2); bell.h(0); bell.cx(0, 1)
psi = Statevector(bell)
print(psi)                         # [0.707+0j, 0, 0, 0.707+0j]

rho0 = partial_trace(psi, [1])    # trace out qubit 1
print(entropy(rho0))              # 1.0 — maximally mixed, i.e. maximally entangled
```

That entropy value is the point. Qubit 0 alone carries **one full bit of entropy** — no information at all — while the pair is in a perfectly known pure state. All the information lives in the correlation.

Connecting to IBM Quantum

```
from qiskit_ibm_runtime import QiskitRuntimeService, SamplerV2

service = QiskitRuntimeService()              # reads saved credentials
backend = service.least_busy(operational=True, simulator=False)

# Transpilation is mandatory, not cosmetic: it maps your abstract circuit
# onto the device's native gate set and physical qubit connectivity.
qc_hw = transpile(qc, backend, optimization_level=3)

job = SamplerV2(mode=backend).run([qc_hw], shots=4096)
counts = job.result()[0].data.c.get_counts()
print(counts)
# {'00': 1902, '11': 1887, '01': 154, '10': 153}  — the off-diagonal terms are noise
```

Those ~4% wrong outcomes are the subject of Week 13. On an ideal device they are zero; on real hardware they measure your gate and readout fidelity.

> **Analogy boundary — transpilation as compilation.** The parallel is close: lower an abstract program onto a target ISA, optimize, allocate registers. Where it diverges is **connectivity**. Two qubits can only interact if physically coupled, so the transpiler inserts SWAP networks to move states adjacent — and each SWAP is three CNOTs, each with error. A compiler's register allocation costs you spills; a transpiler's routing costs you *fidelity*, and on NISQ hardware routing overhead often dominates the algorithm itself.

Key takeaways

- n qubits span 2^n dimensions; Qiskit orders bitstrings little-endian, qubit 0 rightmost.
- Two-qubit separability test: $ad - bc = 0$. Nonzero means entangled.
- H followed by CNOT produces a Bell state — the canonical entangling circuit.
- Phase kickback puts the phase on the *control*; CNOT is a linear operator, not control flow.
- CNOT plus single-qubit rotations is universal. Transpilation costs fidelity through SWAP routing, not just time.

Self-check

Q1 *(Objective 4)* — Your Bell circuit returns `{'00': 512, '11': 512}` on the simulator and `{'00': 470, '11': 468, '01': 45, '10': 41}` on hardware. A colleague says the hardware "produced a different state." Give the more accurate description, and name the two distinct error sources that produce off-diagonal counts.

Q2 *(Objectives 3, 4)* — Explain why the reduced state of one half of a Bell pair has entropy 1 while the pair has entropy 0, and why this is not a violation of information conservation.

Check your answers

A1 — The hardware did not produce a different pure state; it produced a **mixed state** — a statistical ensemble. The ideal output is the pure $|\Phi^+\rangle$; the device output is approximately $\rho = (1-p)|\Phi^+\rangle\langle\Phi^+| + p\cdot(\text{noise})$, with $p \approx 0.09$ here. That is a different *kind* of object, not merely a different vector, and no state preparation error alone accounts for it.

The two independent sources:

Gate error — imperfect H and CNOT operations. The two-qubit gate dominates: CNOT error rates run ~10^{-3}–10^{-2} on current hardware, roughly an order of magnitude worse than single-qubit gates. Coherent errors (systematic over- or under-rotation) and incoherent errors (decoherence during the gate) both contribute.

Readout error — misclassification at measurement. The discriminator assigns $|0\rangle$ or $|1\rangle$ from a noisy analog signal, and confuses them at ~1–3%. Critically, this error occurs *after* the state is correct, so it is separable from gate error and correctable by different means.

Distinguishing them experimentally: run a circuit with **no gates**, preparing $|00\rangle$ and measuring immediately. Any counts other than `00` are pure readout error. Subtract that baseline and the remainder is gate error. This is the basis of measurement-error mitigation, and it is the first diagnostic to run when hardware results look wrong.

A2 — Both statements are true and describe different systems. The **pair** is in a pure state $|\Phi^+\rangle$ — complete knowledge, von Neumann entropy $S(\rho_{AB}) = 0$. The **single qubit** has $\rho_A = \mathrm{tr}_B(|\Phi^+\rangle\langle\Phi^+|) = I/2$, which is maximally mixed, $S(\rho_A) = 1$ bit.

There is no conservation violation because von Neumann entropy is **not additive across subsystems** for entangled states. Classically, $S(AB) \geq \max(S(A), S(B))$ — a whole is at least as uncertain as its parts. Quantum mechanically that inequality can be violated, and here it is maximally violated: $0 \geq 1$ is false. This is exactly the signature of entanglement, quantified as the *entanglement entropy*.

The resolution is that the information is neither in A nor in B. It is in the **correlation**, which the partial trace discards by construction. Tracing out B is the formal statement "I have no access to B," and under that restriction A genuinely is random — not as a description of your ignorance about a hidden value, but as a complete description of the accessible system.

The provenance framing: it is a record whose every field reads as noise, yet which is bit-exact against a counterpart record held elsewhere. The integrity lives in the joint object, and querying either half alone destroys the ability to see it.